

Concept 3 | Introduction to Logarithms and Their Properties

English Student Edition • Focused formulas and guided practice

Formula opener

Definition

$$\log_b a = c \iff b^c = a$$

Product

$$\log_b(xy) = \log_b x + \log_b y$$

Power

$$\log_b(x^k) = k \log_b x$$

Classified Practice

Introduction to Logarithms and Their Properties

Q001 Written

Convert the source forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.

Q002 Written

Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.

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Q003 **Written**

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

Q004 Written

If $\log_5 25 = x$, **evaluate** $\log_{1/2} x$ **without a calculator**.

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Q005 Written

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1$, $\log_5 5$, $\log_2 8$, $\log_2 16$, $\log_5(1/5)$, $\log_7(49^{3/2})$.

Q006 Written

Find the six Try values $\log_5 1$, $\log_3 3$, $\log_4 4$, $\log_3 27$, $\log_2(1/4)$, $\log_5(1/125)$.

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Q007 **Written****Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.**

Q008 Written**Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.**

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Q009 Written

Perform $\log_6 2 + \log_6 3$ **and** $\log_2 30 - \log_2 3 - \log_2 5$.

Q010 Written

Perform the six Try calculations using product, quotient, and power laws, including $4 \log_5 3 - 2 \log_5 15 - \log_5 45$.

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Q011 Written

Perform all 18 source expressions (a)+(r), combining same-base logarithms exactly.

Q012 Written

Simplify the fractional expression $\log_5 75 + 3 \log_5 \sqrt{2} - 2 \log_5 \sqrt{6}$ to one reduced fraction.

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Q013 Written

Given $\log_{10} 2 = a$ and $\log_{10} 3 = b$, express $\log_{10} 24$, $\log_{10}(27/4)$, $\log_{10} 45$, $\log_{10}(24/75)$ in terms of a, b .

Q014 Written

Given $\log_{10} 2 = a$, $\log_{10} 3 = b$, express the four Try arguments 72 , $3^8/10$, 15 , 12 in terms of a , b .

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Q015 Written

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

Q016 Written

If $\log_c 3 = x$, $\log_c 2 = y$, **write** $\log_c 12$, $\log_c(3/8)$, $\log_c \sqrt{6}$, $\log_2 3$, $\log_{3^b}(2^b c)$ **in terms of** x, y .

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Q017 Written

Find $\log_{16} 64$ and $\log_5(1/125)$ by changing to bases 2 and 5.

Q018 Written**Evaluate the five Try logarithms using a suitable common base.**

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Q019 **Written**

Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.

Q020 Written**Find $\log_9(\sqrt{3}/4\sqrt{2})$ exactly.**

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Q021 Written

Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.

Q022 Written**Solve the four Try products of logarithms after making the bases the same.**

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Q023 Written

Simplify all 15 Exercise products and sums of logarithms by common-base conversion.

Q024 Written**Simplify** $\log_x(\sqrt{y}) \log_{y^2}(z^3) \log_{\sqrt{z}}(x^4)$ **for** $x > 1$.

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Q025 Written

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.

Q026 Written

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.

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Q027 **Written**

State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and $1/3$.

Q028 Written**Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.**

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Q029 Written

For $f(x) = \log_5(x - 2) + 1$ and $g(x) = 5^{x-1} + 2$, state their exact geometric relationship and line of symmetry.

Q030 Written

Order the displayed logarithmic sets, including $\log_2 3$, $\log_2 5$, $2\log_2(1/2)$, $\log_{1/2} 3$, $\log_{1/2} 5$.

Classified Practice**Introduction to Logarithms and Their Properties****Q031** Written**Order the four Try sets of logarithms using monotonicity and change of base.**

Q032 Written**Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.**

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Introduction to Logarithms and Their Properties

Q033 Written

If $\log_c(3^2/2) = x$ and $\log_c(9) = y$, write the relation between x and y for $0 < c < 1$ and $c > 1$.

Q034 MCQ $\log_2 32$ equals:

Choose the correct option.

A 2

B 4

C 5

D 32

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Introduction to Logarithms and Their Properties

Q035 MCQ

 $\log_{1/2} 2$ equals:

Choose the correct option.

A -2

B -1

C 1

D 2

Q036 MCQ $\log_5(1/125)$ equals:**Choose the correct option.****A** -3**B** -2**C** 2**D** 3

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Introduction to Logarithms and Their Properties

Q037 MCQ

 $\log_7 49^{3/2}$ equals:

Choose the correct option.

A 1

B 2

C $3/2$ D $7/2$

Q038 MCQ

$\log_a M + \log_a N$ equals:

Choose the correct option.

A $\log_a(M + N)$

B $\log_a(MN)$

C $\log_a(M/N)$

D $\log_a(M - N)$

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Introduction to Logarithms and Their Properties

Q039 MCQ

The coefficient $3 \log_5 2$ becomes:**Choose the correct option.**

A $\log_5(2 + 3)$

B $\log_5 2^3$

C $\log_5(3^2)$

D $3^2 \log_5 2$

Q040 MCQ**If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:****Choose the correct option.**

- A $a + b$
- B $2a + 3b$
- C $3a + b$
- D $3a + 2b$

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Q041 MCQ

$\log_c \sqrt{6}$ in terms of $x = \log_c 3, y = \log_c 2$ is:

Choose the correct option.

A $x + y$

B $(x + y)/2$

C $2x + y$

D $x + 2y$

Q042 MCQ $\log_{16} 64$ equals:

Choose the correct option.

A $1/2$ B $3/2$

C 2

D 4

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Q043 MCQ

Change of base gives $\log_a b =$:**Choose the correct option.**

A $\log_c a / \log_c b$

B $\log_c b / \log_c a$

C $\log_a b / \log_c b$

D $\log_c a \log_c b$

Q044 MCQ $\log_4 16 \cdot \log_9 3$ equals:

Choose the correct option.

A $1/2$

B 1

C 2

D 4

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Introduction to Logarithms and Their Properties

Q045 MCQ

The challenge product of three logarithms equals:**Choose the correct option.**

A 6

B 8

C 12

D 24

Q046 MCQ

The domain of $y = \log_a x$ is:**Choose the correct option.**A all real x B $x \geq 0$ C $x > 0$ D $x \neq 0$

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Q047 MCQ

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

Choose the correct option.

A $x = 0$

B $y = 0$

C $y = x$

D $y = -x$

Q048 MCQ**For $a > 1$ and $M < N$, which is true?****Choose the correct option.**

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal**D** Cannot compare

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Q049 MCQ

For $0 < a < 1$ and $M < N$, which is true?**Choose the correct option.**

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal

D Both are positive

Q050 MCQ

The vertical asymptote of $y = \log_a x$ is:**Choose the correct option.**

A $x = 0$

B $y = 0$

C $x = 1$

D $y = 1$

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Q051 MCQ

A negative coefficient $-2 \log_a M$ can be written as:

Choose the correct option.

- A $\log_a M^{-2}$
- B $\log_a(-2M)$
- C $\log_a(M - 2)$
- D $2 \log_a M$

Q052 MCQ

The base of $\log_a M$ is:**Choose the correct option.**A M B a C p D $1/a$

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Introduction to Logarithms and Their Properties

Q053 MCQ

The argument of $\log_a M$ is:**Choose the correct option.**A a B M C p D 0

Q054 MCQ $\log_2(1/8)$ equals:**Choose the correct option.****A** -3**B** -2**C** 2**D** 3

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Introduction to Logarithms and Their Properties

Q055 MCQ

 $\log_4 4$ equals:

Choose the correct option.

A 0

B 1

C 2

D 4

Q056 MCQ $\log_3 1$ equals:

Choose the correct option.

A -1

B 0

C 1

D 3

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Q057 MCQ

 $\log_a M - \log_a N$ equals:

Choose the correct option.

A $\log_a(MN)$

B $\log_a(M/N)$

C $\log_a(M - N)$

D $\log_a(N/M)$

Q058 MCQ **$2 \log_3 5$ equals:****Choose the correct option.****A** $\log_3 10$ **B** $\log_3 25$ **C** $\log_3 7$ **D** $\log_3(5/2)$

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Q059 MCQ

If $a = \log_{10} 2$, $b = \log_{10} 3$, $\log_{10} 12$ is:

Choose the correct option.

A $a + b$

B $2a + b$

C $a + 2b$

D $2a + 2b$

Q060 MCQ

$\log_{10} 5$ in terms of $a = \log_{10} 2$ is:

Choose the correct option.

A a

B $1 - a$

C $1 + a$

D $-a$

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Q061 MCQ

 $\log_8 32$ equals:

Choose the correct option.

A $2/3$ B $3/2$ C $5/3$

D 3

Q062 MCQ

$\log_{25} 125$ equals:

Choose the correct option.

A $1/2$

B $3/2$

C 2

D $5/2$

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Introduction to Logarithms and Their Properties

Q063 MCQ

 $\log_a b \cdot \log_b a$ equals:

Choose the correct option.

A 0

B 1

C 2

D $\log_a b$

Q064 MCQ

$\log_2 8 \cdot \log_8 2$ equals:

Choose the correct option.

A $1/3$

B 1

C 3

D 8

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Q065 MCQ

The best common base for 16 and 64 is:**Choose the correct option.**

A 2

B 4

C 8

D 10

Q066 MCQ

The range of $y = \log_a x$ is:**Choose the correct option.**A $x > 0$ B $\mathbb{R}$ C $[0, \infty)$ D $y > 0$

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Introduction to Logarithms and Their Properties

Quiz MCQ 1 **Concept Quiz · MCQ**

The logarithmic equivalent of $a^p = M$ is:

Choose the correct option.

A $\log_a M = p$

B $\log_p M = a$

C $\log_M a = p$

D $\log_a p = M$

Quiz MCQ 2 Concept Quiz · MCQ

$\log_2 8 + \log_2 4$ equals:

Choose the correct option.

A 2

B 3

C 5

D 12

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Introduction to Logarithms and Their Properties

Quiz MCQ 3 **Concept Quiz · MCQ**

$\log_{16} 64$ equals:

Choose the correct option.

A $1/2$

B $3/2$

C 2

D 4

Quiz MCQ 4 **Concept Quiz · MCQ**

For $a > 1$, $M < N$ implies:

Choose the correct option.

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C equality

D no relation

Classified Practice

Introduction to Logarithms and Their Properties

Quiz WRITTEN 5 Concept Quiz · Written

Find $\log_3(1/27)$.

Quiz WRITTEN 6 Concept Quiz · Written**Simplify** $\log_5 25 + \log_5 5$.

Classified Practice

Introduction to Logarithms and Their Properties

Quiz WRITTEN 7 **Concept Quiz · Written**

Find $\log_{16} 64$ using change of base.

Quiz WRITTEN 8 Concept Quiz · Written

State the domain and range of $y = \log_a x$.

Answer key

Use the question number shown on each page.

- Q001** $\log_2 32 = 5$; $3^2 = 9$
- Q002** $\log_2 64 = 6$; $\log_4(1/16) = -2$, with the same swap rule for the remaining items.
- Q003** Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.
- Q004** $x = 2$, so $\log_{1/2} 2 = -1$.
- Q005** Values: 0, 1, 3, 4, -1, 3/2.
- Q006** 0, 1, 1, 3, -2, -3.
- Q007** Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.
- Q008** 2/3.
- Q009** 1 and 1.
- Q010** Each expression reduces to one logarithm and then an integer; the displayed worked example gives -3.
- Q011** Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.
- Q012** 1.
- Q013** $3a + b$, $2 - 2a + 3b$, $2a + 1 - a$, $b + 2 - 2a$.
- Q014** Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.
- Q015** Each result is a linear combination of $a, b, 1$ obtained from prime factors.
- Q016** $x + 2y$, $x - 3y$, $(x + y)/2$, x/y , $(1 + by)/(bx)$.
- Q017** 3/2 and -3.
- Q018** Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.
- Q019** All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.
- Q020** $\frac{\log_2 3 - 5/2}{2 \log_2 3}$.
- Q021** Use reciprocal and power identities; the worked values are exact constants.
- Q022** Each product collapses to a rational number after change of base and cancellation.
- Q023** Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.
- Q024** 12.
- Q025** They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.
- Q026** Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.
- Q027** Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.
- Q028** Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.
- Q029** They are inverse-reflected curves with symmetry line $y = x + 1$.
- Q030** Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.
- Q031** Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.
- Q032** Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.
- Q033** Use the same argument comparison; the relation reverses when the base crosses 1.
- Q040** C
- Q041** B
- Q042** B
- Q043** B
- Q044** A
- Q045** C
- Q046** C
- Q047** C
- Q048** B
- Q049** A
- Q050** A
- Q051** A
- Q052** B
- Q053** B
- Q054** A
- Q055** B
- Q056** B
- Q057** B
- Q058** B
- Q059** B
- Q060** B
- Q061** B
- Q062** B
- Q063** B
- Q064** B
- Q065** A

Q066 B**Quiz MCQ 1 A****Quiz MCQ 2 C****Quiz MCQ 3 B****Quiz MCQ 4 B****Quiz WRITTEN 5 -3** **Quiz WRITTEN 6 3****Quiz WRITTEN 7 $3/2$** **Quiz WRITTEN 8** Domain $x > 0$; range all real numbers